

- (d) KHMS schools have one or more annexes. Each teacher is assigned to only one annex. Each teacher may coordinate one or more classes. Each class is coordinated by only one teacher. Also, each teacher may submit one or more assessments but each assessment is submitted by only one teacher. Each class may have one or more assessments and an assessment may be for more than one class. Create an ERD for this narrative. (10 marks)

CSC317 TEST 1(15)
2022/2023
TIME: 35 MINS
ANSWER ALL QUESTIONS

Analy.

1. How does an analyst solve problems? [4marks]
2. State the sequence of an analyst approach to solving problems with the aid of a diagram? [4 marks]
3. Briefly mention the required skills of systems analyst? *7 Admin* [2marks]
4. List the SDLC phases? [2.5marks]
5. In confirming project feasibility mention 4 factors that are to be considered during this stage? [1mark]
6. Mention any three (3) goals of requirement analysis? [1.5 marks]

is a new patient. If it is a change or cancellation, find the call or patient to the account's office. If it is a change or cancellation, find

COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.SC EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

DEPARTMENT: COMPUTER & INFO. SCIENCES

SESSION: 2022/2023

SEMESTER: ALPHA

COURSE CODE: CSC317

CREDIT UNIT: 2

COURSE TITLE: SYSTEM ANALYSIS AND DESIGN

INSTRUCTION: Answer ANY three (3) Questions

TIME: 2 HOURS

QUESTION ONE

- (a) Briefly describe the following Object-Oriented Design concepts with relevant examples:
(i) Polymorphism, (ii) Inheritance, (iii) Composition and (iv) Aggregation (4.5 marks)
- (b) State two (2) advantages and one disadvantage of inheritance (3 marks)
- (c) With illustrations, describe the Martins ERD Notation style. (6 marks)
- (d) KHMS schools have one or more annexe. Each annexe may have one or more teachers, but a teacher is assigned to only one annexe. Each teacher may coordinate one or more classes but each class is coordinated by only one teacher. Also, each teacher may submit one or more class assessments but each assessment is submitted by only one teacher. Each class may have many assessments and an assessment may be for more than one class. Create an ERD for this narrative. (10 marks)

QUESTION TWO

- (a) Discuss briefly the major phases in the System Development Life cycle and identify the deliverables in each phase. (9 marks)
- (b) You are to develop a sales management system that allows sales representative to record orders placed by customers, checkout orders, and also allows the storekeeper to keep inventory of product in the warehouse. A customer can place more than one order however each order must be placed by one customer. Each sales representative serves zero to many customers per time, however each customer is served by only one sales representative. Each order contains one or more products and zero or more products are stored in only one warehouse.
- i) Draw an entity relationship diagram for this scenario. (5 marks)
- ii) Sketch a Use Case diagram to depict sales representative and storekeeper functions [NOTE: recording order includes updating order, and checkout extends edit order details. Keep inventory includes update products details.] (5 marks)
- iii) Draw a sequence diagram for the sales representative's interaction with the system given that a customer's order can have up to 12 products. (4.5 marks)

QUESTION THREE

- (a) Patients call or come to the hospital to make new appointments as well as change or cancel appointments. Check to make sure the patient is a current patient and has no unpaid bills. If this is a new patient, perform "Add patient" before continuing. If the patient has unpaid bills, forward the call or patient to the account's office. If it is a change or cancellation, find the current

appointment in the appointment file and cancel it. Match the patient's desired appointment times with available dates and times and schedule new appointment. Draw Process 0 and Level 1 Data Flow Diagrams. (10 marks)

- (b) Identify and illustrate common mistakes to avoid in Process modelling. (6 marks)
(c) Distinguish between tools, technique and methodologies with relevant examples. (7.5 marks)

QUESTION FOUR

- (a) Hebron Consultancy has hired your company to develop an information system for its operations. Hebron Consultancy desires speedy delivery of the system and active users engagement in the development process. As the head of the project team:
- Discuss three (3) fact finding methods you intent to use in gathering requirement for the proposed system. (7.5 marks)
 - Give details of your overall strategy for the fact finding exercise. (5 marks)
- (b) You are to choose between V-shape, Prototyping and Water fall as the development methodology for the proposed system. State the methodology of your choice with reasons. Give a schematic view of the methodology and state its advantages and disadvantages. (8 marks)
- (c) State three (3) advantages and disadvantages of Agile development methodology. (3 marks)

QUESTION FIVE

- (a) How does an analyst solves problem? Explain your answer with tangible reasons. (5 marks)
- (b) As the head of an interview panel to recruit system analyst for your company, explain in detail the business and people knowledge expected for the candidate to be successful. (6 marks)
- (c) Name three tools used in system development and give at least two examples for each (4.5 marks)
- (d) Draw the Class diagram showing the Relationships and Multiplicities for the following analysis information. Also specify at least two attributes and one method for each class:
- An Order is made-up of three OrderItems
 - An Order can either be a SpecialOrder or a NormalOrder
 - A Customer can place at most 20 Students per time
- (8 marks)

